Candidate's name ……………………………………………… CTG ……………

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS 2018

CHEMISTRY — 9729/4
HIGHER 2 — **6 August 2018**

2 hours 30 minutes

Paper 4 Practical Test

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces provided on this cover page.

Give details of the practical shift and laboratory, where appropriate, in the box provided.

Write in dark blue or black pen.
You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

Qualitative Analysis Notes are printed on pages 18 and 19.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	/41
2	/14
Total	
	/ 55

Answer **all** the questions in the spaces provided.

1 Determination of concentration of sulfuric acid

Both group 1 carbonates and hydroxides are bases which neutralise sulfuric acid as shown by the equations below.

$M_2CO_3 + H_2SO_4 \rightarrow M_2SO_4 + H_2O + CO_2$
$2MOH + H_2SO_4 \rightarrow M_2SO_4 + 2H_2O$

FA 1 is dilute solution of sulfuric acid, H_2SO_4
FA 2 is 0.550 mol dm^{-3} sodium hydroxide, NaOH
FA 3 is anhydrous powder of an unknown group 1 carbonate

In this question, you will perform an experiment to determine the concentration of the sulfuric acid solution by means of a graphical method and hence, determine the relative atomic mass of an unknown group 1 element.

You will first prepare a standard solution of the group 1 carbonate, **FA 4**, using its anhydrous powder, **FA 3**.

You will then add different volumes of **FA 4** to identical samples of dilute sulfuric acid, **FA 1**. In each case, the amount of the group 1 carbonate solution you add will only partially neutralise the sulfuric acid.

You will then complete the neutralisation of each mixture by titration with sodium hydroxide, **FA 2**.

In **1(c)**, you will use your volumes of **FA 2** and **FA 4** to plot a graph which will enable you to determine the concentration of the sulfuric acid solution and hence, determine the relative atomic mass of the unknown group 1 element.

(a) Preparation of **FA 4**

1. Weigh accurately about 10.0g of **FA 3** into an empty weighing bottle. Record all weighings in the space provided. Make certain that your recorded results show the precision of your working.
2. Transfer the contents of the weighing bottle into a 250 cm^3 beaker and rinse the weighing bottle with deionised water and transfer the rinsing into the beaker. Repeat the rinsing to ensure all **FA 3** has been transferred.
3. Add 100 cm^3 of deionised water to dissolve **FA 3**.
4. Transfer **all** of the solution to a graduated flask and make up the solution to 250 cm^3 with deionised water and mix thoroughly.

Results

[1]

(b) Preparation and titration of the reaction mixture

Notes: You will perform each titration **<u>once</u>** only. Great care must be taken that you do not overshoot the end-point.

You should aim not to exceed 25 minutes for this experiment.

1. Fill the burette labelled '**FA 2**' with **FA 2**.
2. Fill the burette labelled '**FA 4**' with **FA 4**.
3. Pipette 25.0 cm^3 of **FA 1** into a 250 cm^3 conical flask.
4. Add 6.00 cm^3 of **FA 4** into the same conical flask from step 3.
5. Thoroughly swirl the mixture.
6. Add 2 to 3 drops of methyl orange indicator to the conical flask.
7. Titrate the mixture in the conical flask with **FA 2** until an orange colour is obtained.
8. Repeat steps 3 to 7 until four more aliquots using the following volumes of 9.50 cm^3, 13.00 cm^3, 17.00 cm^3 and 20.00 cm^3 of **FA 4** have been titrated. Record your results in the space provided. Make certain that your recorded results show the precision of your working.

Results

[2]

(c) Plot a graph of the volumes of **FA 2** added, on the *y*-axis, against the volumes of **FA 4** added on the *x*-axis on the grid in **Fig 1.1.**

You should choose scales for the axes which allow you to determine by extrapolation

- the volume of **FA 4** required, $\mathbf{V_{max}}$**(FA 4)**, to completely react with 25.0 cm^3 of **FA 1** if no **FA 2** is added;
- the volume of **FA 2** required, $\mathbf{V_{max}}$**(FA 2)**, to completely react with 25.0 cm^3 of **FA 1** if no **FA 4** is added.

Draw the line of best fit, taking into account all your plotted points. Hence obtain values for $\mathbf{V_{max}}$**(FA 4)** and $\mathbf{V_{max}}$**(FA 2)**.

Fig 1.1

$\mathbf{V_{max}}$**(FA 4) =……………………….** $\mathbf{V_{max}}$**(FA 2) = …………………………..**

[7]

(d) Suggest an explanation on the direction of the slope of your graph.

[1]

Calculations

(e) **(i)** Using appropriate data from your graph in **(c)**, calculate the concentration of sulfuric acid in **FA 1**.

[2]

(ii) Using your answer in **(e)(i)** and appropriate data from your graph in **(c)**, calculate the concentration of group 1 carbonate in **FA 4**.

[2]

(iii) Hence, calculate the relative atomic mass of M.
[A_r: O = 16.0; C = 12.0]

[2]

(f) A student performs this experiment. Unknowingly, he uses a sample of **FA 3** which is slightly damp. State and explain the effect on the relative atomic mass of M calculated in **e(iii)**.

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.... [2]

Turn over

(g) Determination of concentration of sulfuric acid using gravimetric analysis

A student performed an experiment to determine the concentration of sulfuric acid using its density. The student weighed the mass of an empty 250 cm^3 volumetric flask and recorded the mass Table 1.1. He used a pipette to transfer 25.0 cm^3 of sulfuric acid, **FA 1** into the flask and then re-weigh it. He then recorded the mass of the flask and **FA 1** in Table 1.1.

Table 1.1

Mass of flask and **FA 1** / g	122.22
Mass of flask / g	96.62
Mass of **FA 1**/ g	

(i) Calculate the mass and hence the density of sulfuric acid in **FA 1** in g cm^{-3}. Leave your calculated density to 3 decimal places.

[2]

(ii) Use **Fig. 1.2** and your answer in **g(i)**, deduce the concentration of sulfuric acid in **FA 1** in mol dm^{-3}.

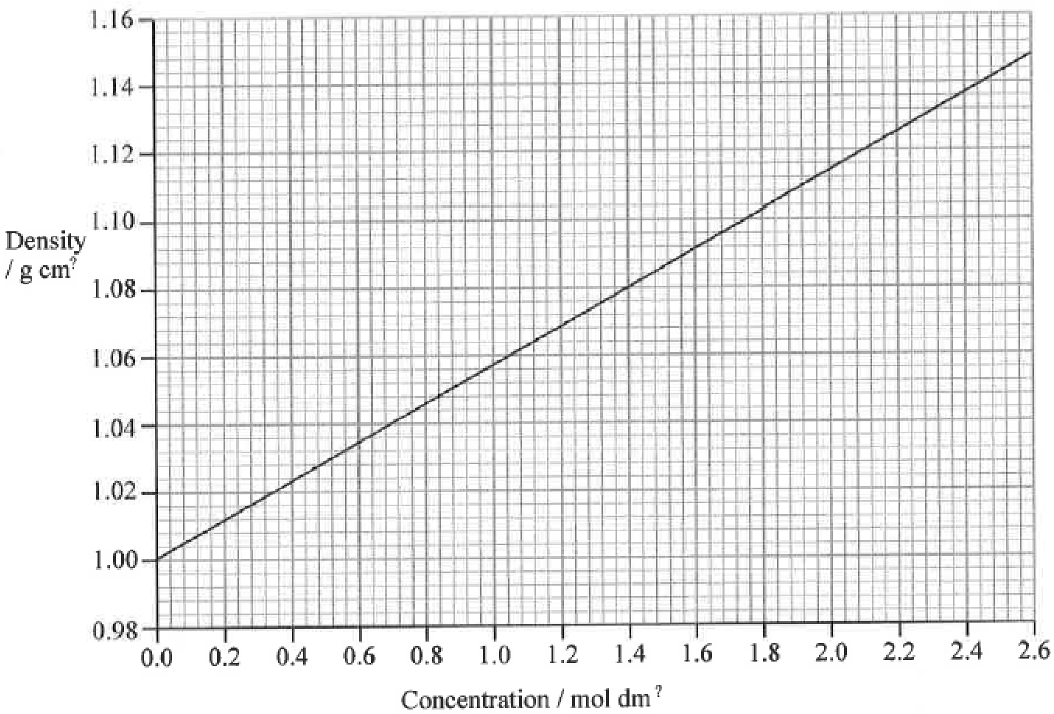

Fig. 1.2

concentration of sulfuric acid in **FA 1** = **…………………………**

[1]

(h) Determination of concentration of sulfuric acid using thermochemistry

The neutralisation between a base and sulfuric acid is exothermic. Heat released is sufficient to allow the maximum temperature, ΔT_{max}, to be determined within a few minutes.

In this question, you are to perform an experiment to determine a value, ΔT_{max}. You will then use your value, ΔT_{max} to determine the concentration of sulfuric acid.

You will carry out the following instructions to obtain a value of ΔT_{max}.

1. Use a 25.0 cm^3 pipette, transfer **FA 1** into a polystyrene cup held securely in a beaker. Record the temperature to 1 decimal place in the table provided.
2. Use a measuring cylinder, measure 100 cm^3 of **FA 2**. Record the temperature to 1 decimal place. Calculate the weighted average temperature, T_{ave} of T_{FA1} and T_{FA2} to 1 decimal place.
3. Add **FA 2** carefully and quickly to **FA 1** in the cup, stir and record the maximum temperature reached, again to 1 decimal place.

Weighted average temperature, T_{ave} = $\dfrac{(V_{FA1} \text{ X } T_{FA1}) + (V_{FA2} \text{ X } T_{FA2})}{V_{FA1} + V_{FA2}}$

Temperature of **FA 1** before mixing, T_{FA1} / ^oC	
Temperature of **FA 2** before mixing, T_{FA2} / ^oC	
Weighted average temperature, T_{ave} / ^oC	
Maximum temperature reached, T_{max} / ^oC	
ΔT_{max} / ^oC	

[2]

(i) Use the equation below to find the concentration of the acid:

Concentration = 0.161 × ΔT_{max}

Concentration = ……………………………….

[1]

(ii) Instead of adding **FA 2** to **FA 1** to determine ΔT_{max}, the student added barium hydroxide, $Ba(OH)_2$ of the same concentration and volume. Explain the impact on the value of ΔT_{max}.

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.... [2]

Turn over

(i) Planning

The exothermic reaction between an acid and metal hydroxide can be used to determine the equivalence-point of a neutralisation reaction without the use of an indicator. This process is known as *thermometric titration* and can be used to calculate the concentration of an acid solution.

The procedure from **1(h)** could be further improved to obtain a more accurate value for the concentration of sulfuric acid in **FA 1**.

Plotting ΔT_{max} against volumes of **FA 2** gives two best-fit lines. One line is drawn using data before the equivalence-point and the second line using the remaining data. These lines are then extrapolated until they intersect.

(i) Plan an investigation, based on the information above, to determine the equivalence-point for the neutralisation between **FA 1** and **FA 2** and hence the concentration of sulfuric acid in **FA 1**.

You may assume you are provided with
- 250 cm^3 solution of dilute sulfuric acid, **FA 1**,
- 250 cm^3 of 0.550 mol dm^{-3} sodium hydroxide, NaOH, **FA 2**,
- graph paper,
- the equipment normally found in the school or college laboratory.

In your plan you should include brief details of
- the apparatus you would use, bearing in mind the levels of precision they offer,
- the procedure that you would follow and the measurements that you would take,
- how you would recognise the equivalence-point had been passed,
- provide a sketch of the graph that you expect to obtain,
- how you would determine the concentration of sulfuric acid,
- state a safety precaution and suggest how do you minimise the risks involved.

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[8]

(ii) State and explain one consideration to be taken when deciding the minimum volume of **FA 1** to be used.

.... [1]

(iii) Describe how you would use your graph in **1i(i)** to determine the enthalpy change of neutralisation between **FA 1** and **FA 2.**

.... [2]

(iv) How, if at all, will ΔT_{max} vary if you use half the volumes of the original **FA 1** and **FA 2**. Explain your answer.

.... [2]

(v) Provide a reason why this method is preferred to that in **1(h)**.

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.... [1]

[Total: 41]

Turn over

2 Investigation of some chemical reactions involving metal ions

FA 5 is a solid metal carbonate with the formula, XCO_3.
FA 6 is an aqueous solution containing one cation and one anion.

(a) Perform the tests described in **Table 2.1**, and record your observations in the table. Test and identify any gases evolved.

Table 2.1

	tests	observations
1	Place a spatula of solid **FA 5** in a test-tube. Hold the test-tube in a holder. Heat the test-tube strongly until no further changes are seen. Keep the residue for test 2.	
2	To the residue from test 1, add sufficient amount of **FA 1** to dissolve the residue completely. Filter the mixture if necessary. Keep the filtrate for tests 3 and 4.	
3	Add ten drops of the solution obtained from test 2 to a test-tube. Add aqueous NH_3 to this test-tube, until no further change.	

4	Add ten drops of the solution from test 2 to a test-tube. Then add 5 drops of aqueous potassium iodide and shake the mixture thoroughly. To the mixture, add 5 drops of aqueous sodium thiosulfate.	
5	Add a drop of **FA 6** solution onto a universal indicator paper.	
6	Add 1cm depth of **FA 6** to a test tube. Then add a spatula of **FA 3** to the test tube.	
7	To 1 cm depth of **FA 6**, add 10 drops of nitric acid followed by 10 drops of aqueous barium chloride.	

[5]

(b) Consider your observation in **Table 2.1**.

(i) Suggest the identities of the cations in **FA 5** and **FA 6** and the anion in **FA 6**.

Cation in **FA 5** : ……………………………………………………………………..

Cation in **FA 6** : ……………………………………………………………………..

Anion in **FA 6** : ……………………………………………………………………….

[2]

(ii) A student carried out test 1 using **FA 3**, it is unlikely for the same observation to be observed for this test. Suggest an explanation to account for this difference.

[2]

(iii) Test 4 in **Table 2.1** shows a redox reaction to take place.

Some relevant redox half-equations are given in Table 2.2.

Table 2.2

half equation	E^θ / V
$X^{2+} + e^- \rightleftharpoons X^+$	+0.15
$I_2 + 2e^- \rightleftharpoons 2I^-$	+0.54

Using the information in **Table 2.2**, calculate E^θ_{cell} for the reaction shown in your observation in test 4. Suggest a reason for the deviation from the prediction of E^θ_{cell} value calculated.

[2]

(iv) Write an equation to account for the observation to test 5 in **Table 2.1**.

[1]

(v) Use your answer to **(iv)** to account for the observations to test 6 in **Table 2.1.**

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.... [2]

[Total: 14]

Turn over

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu^{2+}(aq),	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. insoluble in excess	green ppt. insoluble in excess
iron(III), Fe^{3+}(aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off-white ppt. insoluble in excess	off-white ppt. insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ions	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $Cl^-(aq)$	gives white ppt. with $Ag^+(aq)$ (soluble in $NH_3(aq)$)
bromide, $Br^-(aq)$	gives pale cream ppt. with $Ag^+(aq)$ (partially soluble in $NH_3(aq)$)
iodide, $I^-(aq)$	gives yellow ppt. with $Ag^+(aq)$ (insoluble in $NH_3(aq)$)
nitrate, $NO_3^-(aq)$	NH_3 liberated on heating with $OH^-(aq)$ and Al foil
nitrite, $NO_2^-(aq)$	NH_3 liberated on heating with $OH^-(aq)$ and Al foil; NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, $SO_4^{2-}(aq)$	gives white ppt. with $Ba^{2+}(aq)$ (insoluble in excess dilute strong acids)
sulfite, $SO_3^{2-}(aq)$	SO_2 liberated with dilute acids; gives white ppt. with $Ba^{2+}(aq)$ (soluble in dilute strong acids)

(c) Test for gases

ions	*reaction*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple

END OF PAPER 4